HARSH BAJANIA

Electronics & Computer Science Student

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ABOUT ME

Dedicated and adaptable Electronics and Computer Science student at Shah & Anchor Kutchhi Engineering College with a strong passion for learning new technologies. Eager to apply foundational programming skills and a flexible mindset to contribute to innovative projects. A creative problem-solver who balances technical pursuits with a love for music.

EDUCATION

B.Tech in Electronics & Computer Science — Sep 2024 – Aug 2028

Shah & Anchor Kutchhi Engineering College, Mumbai FY

CGPA: 8.00

Grade 12 – HSC (Science) — Completed 2024

Champions Science Junior College Percentage:

71.67%

Grade 10 – IGCSE — Completed 2022

The Universal School, Ghatkopar Percentage: 84%

EXPERIENCES

Data Science with Power BI – Skill Development Program at Shah & Anchor Kutchhi Engineering College 10-Day Internship Program

- Gained hands-on experience in Python programming and essential data science libraries.
- Studied machine learning algorithms and their practical applications.
- Built Power BI dashboards and data models for business analytics.
- Completed a final project expo showcasing practical data science applications.

QUERYCRAFT: The Art of Database Engineering — Dec 8–12, 2025

Organized by IEI-SAKEC in collaboration with ECS & CM Departments, Shah & Anchor Kutchhi Engineering College

- Completed a 5-day intensive database engineering workshop covering SQL fundamentals, data manipulation, advanced logic, and automation.
- Integrated Python with databases on Day 4 through a hands-on mini project.
- Engaged in research and publication activities on the final day.
- Received an authorized certification from MITU Skillologies upon completing the project and clearing the online assessment.

PROJECTS

Automated Smart Water Irrigation System MATLAB, Simulink, SolidWorks

- Designed a smart irrigation robot in Simulink to reduce water waste using sensor-guided automation.
- Implemented a 3-DOF robotic arm with Inverse Kinematics to autonomously target dry spots.
- Achieved positioning accuracy of <3.0 mm and a 0.5-second system response time.

Pocket Calculator – (IC 7476 & 7404) Arduino IDE, Arduino C++

- Led software debugging for an Arduino-based calculator, resolving critical SRAM crashes for system stability.
- Developed a state-machine-driven application with an animated OLED UI and comprehensive error handling.

AI-Driven Big Data Analytics Framework C, VSCode

- Designed a 4-phase theoretical architecture for an AI-driven big data processing framework.
- Implemented the foundational data acquisition phase in C using a dynamic linked list-based Queue for FIFO data stream management.

Wine Quality Prediction & Visualization Python, Power BI, Jupyter Notebook

- Developed a wine quality prediction model using physicochemical attributes such as alcohol percentage and age.
- Applied machine learning techniques including predictive modeling, feature engineering, and EDA.
- Built an interactive Power BI dashboard enabling stakeholders to filter trends and identify key quality drivers.

TECHNICAL SKILLS

Programming Languages: C, Java, Python, HTML, CSS, JavaScript
Data & BI Tools: Power BI, Pandas, Jupyter Notebook, Google Colab
Engineering Software: MATLAB/Simulink, SolidWorks, AutoCAD, Arduino IDE
Databases: MongoDB
Productivity & IDEs: VS Code, Verilog (HDL), Microsoft Word, Excel, PowerPoint, Outlook
Operating Systems: Windows, macOS

CERTIFICATIONS

- Java Programming – Spoken Tutorial, IIT Bombay (NMEICT, Govt. of India): OOP, exception handling, GUI development.
- HTML Training – Spoken Tutorial, IIT Bombay (NMEICT, Govt. of India): Semantic structure, forms, media embedding.
- MongoDB Schema Patterns & Antipatterns – MongoDB, Inc.: Scalable data modelling and schema design best practices.
- MongoDB Aggregation Framework – MongoDB, Inc.: Complex data transformation, querying, filtering, and grouping.
- JavaScript Essentials 1 – Cisco Networking Academy & JS Institute, SAKEC: DOM manipulation, event handling, and interactive UI development.

INTERESTS & EXTRACURRICULARS

- **ACM (Association of Computer Machinery) Cell - Newsletter Head : Led** the documentation strategy for all ACM events, coordinating with event heads and writers to ensure accurate and timely newsletter publications.
- Tech Enthusiast – actively explores emerging technologies and self-driven learning.